

WAY4TM

Basic Host-to-Host Protocol

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Overview

This document describes the WA Y 4 host-to-host protocol based on the ISO 8583-87 standard. The WA Y 4 host-to-host protocol can be used to provide authorisation and information services to card processing hosts or transaction concentrators.

The document presents the basic subset of the WA Y 4 host-to-host protocol supporting Network Management, Authorisation, Authorisation Advice, Reversal, and Balance Inquiry.

Chapter 1. Transport Layer

The WA Y 4 host-to-host supports the following transport protocols:

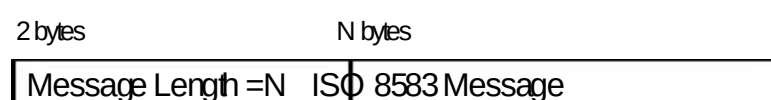
- TCP/IP
- X.25

TCP/IP Transport

TCP/IP is a point to point communication protocol with a delivery and data sequence guarantee. The connection party which initiates a connection is a TCP/IP *client*. The connection party which accepts a connection is a TCP/IP *server*. A TCP/IP client establishes a connection identifying a target application with an IP address of the host and a port number within that host. If the connection is broken by any reason it should be re-established by the client application with a recommended delay of 1 second.

In a TCP/IP connection, the WA Y 4 host can act either as a client or as a server depending on the WA Y 4 host configuration.

TCP/IP uses sessions for data transmission. Each session is a bi-directional data stream. The WA Y 4 host-to-host protocol uses a single TCP/IP session to transfer data between hosts in both directions. A continuous TCP/IP data stream is split into frames. Each ISO 8583 message is sent in a separate frame. A frame consists of a 2-byte header and a message body. A header contains the length of the following message. The high byte of value is transmitted first, and the low byte of value is transmitted second.



The WA Y 4 host-to-host protocol defines the message consistency control. The waiting time for the message data is limited with the T1 timer (the recommended value is 5 seconds) started on receiving the first byte of the frame header. If the message of the length specified in the frame header is not received during the T1 timer period, the connection is broken and should be re-established by the client host.

To check a TCP/IP session state, hosts may use zero-length packets. It is a 2-byte frame header with the '0' value that is not followed by any data. If a host receives this test packet, it should discard it without any additional processing.

X.25 Transport

The WA Y 4 host-to-host connection supports the X.25 data communications protocol. The communications protocol is based on the CCITT Recommendation X.25. Some restrictions are applied to the use of the X.25 transport with the WA Y 4 host-to-host protocol:

- Switched Virtual Circuits (SVCs) must be used for the connection.
- If several WA Y 4 host-to-host connections are established with one host, the combination of the Called Address, the Calling Address and the User Data must be unique to identify the connection.
- If a connection is broken by any reason, it should be re-established by the calling party with the recommended delay of 1 second
- The host must be capable of sending and receiving messages using the M-bit (More bit) technology. M-bit is used to transmit application messages that exceed an X.25 data packet length. Long messages should be split into two or more X.25 data packets. Every X.25 data packet except the last one must have the M-bit set.

Chapter 2. Application Layer

ISO 8583 Basics

The establishment and termination of an application session and all data transfers within a session are achieved by means of 'application messages'. All messages consist of three components and have the following structure:

MESSAGE TYPE IDENTIFIER	BIT MAP(S)	DATA ELEMENTS
-------------------------	------------	---------------

Message Type Identifier (MTID)

Each message starts with a Message Type Identifier (MTID), which is a four-digit numeric field identifying the message version, class, function, and originator, as defined by ISO 8583. The MTID field consists of 4 ASCII coded bytes with MTID value. The online WA Y 4 host-to-host interface uses the following values:

- First digit – Version Number:
0: ISO 8583 1987
1: ISO 8583 1993
9: Reserved for private use
- Second digit – Message Class:
1: Authorisation
2: Financial
3: File Action
4: Reversal/Chargeback
5: Reconciliation
6: Administrative
7: Fee collection
8: Network Management
- Third digit – Message Function:
0: Request
1: Request Response
2: Advice
3: Advice Response
4: Notification
- Fourth digit – Transaction Originator:
0: Acquirer
1: Acquirer Repeat

- 2: Card Issuer
- 3: Card Issuer Repeat
- 4: Other
- 5: Other Repeat

Bit Maps

The second component of a message is one, two or three "bit maps" referred to as the "Primary Bit Map", "Secondary Bit Map" and "Third Bit Map". Each consists of 8 bytes and is interpreted as 64 individual bits. For the Primary Bit Map, the bits are numbered from the left starting with "1", up to "64". Each bit signifies the presence (1) or the absence (0) in the message of the data element associated with that particular bit, according to its position (i.e. number) within the bit map. The Primary Bit Map must always be present. The first bit of the bit map indicates the presence (1) or the absence (0) of the next bit map. For the Secondary Bit Map, the bits are numbered starting with "65", up to "128". The Secondary Bit Map must be present in a message if the message contains a data element numbered 65 or higher. For the Third Bit Map, the bits are numbered starting with "129", up to "192". The Third Bit Map must be present in a message if the message contains a data element numbered 129 or higher.

Data Elements

The final component of a message consists of a number of "data elements". Data elements may be of fixed or variable length. No delimiters are used between data elements. Their order and presence is indexed by the associated bit map(s). Variable length data starts with length specifier indicating the length of that particular data element. Each data element is characterised by:

- its name and data element number
- its bit map position (as given by the data element number)
- its attributes, expressed in terms of alpha, numeric, special characters, or binary
- its length, expressed as a number of bytes or characters
- its contents and format
- its intended use

See "Data Elements" for a full description of the format, content and use of all data elements and private data sub-elements that may be used in application messages.

Message Formats

Interpreting Tables in this Chapter

Format Table Example

DE	DE Name	0100	0110
02	Primary Account Number	M	M
03	Processing Code	M	M
04	Amount, Transaction	C	C
06	Amount, Cardholder Billing	C	C
07	Transmission Date and Time	M	M
14	Expiration Date	C	
18	Merchant's Type	M	
20	PAN Extended, Country Code	C	C

Column Meanings

- DE** – the sequence number of the Data Element.
- DE Name** – the name of the Data Element according to the ISO 8583 standard.
- MTID** – the condition of the Data Element presence for messages with the specified MTID (see "DE Presence Conditions").

DE Presence Conditions

- C** – Conditional. The field/value is present in the message under certain conditions explained in Data Element definitions.
 - M** – Mandatory. The field/value must be present in the message.
 - O** – Optional. The field/value presence in the message is up to the message initiator/responder.
 - Grey** – Echo. For response messages, the field must be the same as in the corresponding request. For derivative message, the value must be the same as in the corresponding original message.
- Fields not listed in the Format Table or having blank field conditions might be present in a message. If such fields are present, their structure must conform to the ISO8583 field description. If it does not, the message might be discarded. The value of these fields should be ignored without any impact on message processing.

Explanation Examples

- See "Format Table Example".
- Field 02: 1) In 0100 messages, this field must be present. 2) In 0110 messages, this field must be present and equal to the value of the same field in the 0100 message.
- Field 04: 1) In 0100 messages, this field may be present or absent depending on certain conditions. 2) If the field is present in an 0100 message, it must be present in the corresponding 0110 message and have the same value. 3) If the field is absent in an 0100 message, it must be omitted in the 0110 response as well.
- Field 18: 1) In 0100 messages, this field must be present. 2) In 0110 messages, this field should not be present. 3) If the field present in an 0110 message, it must have a correct structure, but its value must be ignored.

Network Messages (0800/0810)

Network management messages are messages that (1) establish and change a network status of an application session and (2) conduct an echo test.

Field Number and Name	0800	0810	
7	Transmission Date and Time M	M	
11	System Trace Audit Number M	M	
39	Response Code		M
70	Network Management Information Code M	M	

Authorisation Messages (0100/0110)

Field Number and Name	0100	0110	
02	Primary Account Number (PAN)	M	M
03	Processing Code	M	M
04	Amount, Transaction	M	M
06	Amount, Cardholder Billing	C	C
07	Transmission Date and Time	M	M
10	Conversion Rate, Cardholder Billing	C	C
11	System Trace Audit Number	M	M
12	Time, Local Transaction	M ¹	
13	Date, Local Transaction	M ²	
14	Date, Expiration	C	
18	Merchant Type	M	
19	Acquiring Institution Country Code	C	C
20	Primary Account Number Country Code	C	C
22	Point of Service Entry Mode	M	
23	Card Sequence Number	C	C
25	Point of Service Condition Code	C	C
26	Point of Service (POS) PIN Capture Code	C	
28	Amount, Transaction Fee	C	
32	Acquiring Institution Identification Code	M	M
33	Forwarding Institution Identification Code	C	
35	Track-2 Data	C	
37	Retrieval Reference Number	M	M
38	Authorisation Identification Response		C
39	Response Code		M
41	Card Acceptor Terminal Identification	C	C
42	Card Acceptor Identification Code	C	C
43	Card Acceptor Name/Location	C	
45	Track-1 Data	C	
46	Proprietary Field 46	C	C
47	Proprietary Field 47	C	C
48	Proprietary Field 48	C	C
49	Currency Code, Transaction	M	M
51	Currency Code, Cardholder Billing	C	C
52	Personal Identification Number (PIN) Data	C	
53	Security Related Control Information	C	

¹ For transit requests, this field may be omitted.

² For transit requests, this field may be omitted.

54	Additional Amounts		C
55	Smart Card Specific Data	C	C
100	Receiving Institution Identification Code	C	
102	Account Identification-1	C	C
103	Account Identification-2	C	C
104	Transaction Description	C	C
112	Proprietary Field 112	C	C

Balance Inquiry Messages (0100/0110)

Field Number and Name		0100	0110	
02	Primary Account Number (PAN)		M	M
03	Processing Code		M	M
04	Amount, Transaction		M	M
06	Amount, Cardholder Billing		C	C
07	Transmission Date and Time		M	M
10	Conversion Rate, Cardholder Billing		C	C
11	System Trace Audit Number		M	M
12	Time, Local Transaction		M ³	
13	Date, Local Transaction		M ⁴	
14	Date, Expiration		C	
18	Merchant Type		M	
19	Acquiring Institution Country Code		C	C
20	Primary Account Number Country Code		C	C
22	Point of Service Entry Mode		M	
23	Card Sequence Number		C	C
25	Point of Service Condition Code		C	C
26	Point of Service (POS) PIN Capture Code		C	
28	Amount, Transaction Fee		C	
32	Acquiring Institution Identification Code		M	M
33	Forwarding Institution Identification Code		C	
35	Track-2 Data		C	
37	Retrieval Reference Number		M	M
38	Authorisation Identification Response			C
39	Response Code			M
41	Card Acceptor Terminal Identification		C	C
42	Card Acceptor Identification Code		C	C
43	Card Acceptor Name/Location		C	
45	Track-1 Data		C	
46	Proprietary Field 46		C	C
47	Proprietary Field 47		C	C
48	Proprietary Field 48		C	C
49	Currency Code, Transaction		M	M
51	Currency Code, Cardholder Billing		C	C
52	Personal Identification Number (PIN) Data		C	
53	Security Related Control Information		C	
54	Additional Amounts			C
55	Smart Card Specific Data		C	C
102	Account Identification-1	C	C	
103	Account Identification-2	C	C	
104	Transaction Description	C	C	

³ For transit requests, this field may be omitted.

⁴ For transit requests, this field may be omitted.

112

Proprietary Field 112

C

C

Authorisation Advice Messages (0120/0130)

Field Number and Name		0120	0130	
02	Primary Account Number (PAN)		M	M
03	Processing Code		M	M
04	Amount, Transaction		M	M
06	Amount, Cardholder Billing		C	C
07	Transmission Date and Time		M	M
10	Conversion Rate, Cardholder Billing		C	C
11	System Trace Audit Number		M	M
12	Time, Local Transaction		M ⁵	
13	Date, Local Transaction		M ⁶	
14	Date, Expiration		C	
18	Merchant Type		M	
19	Acquiring Institution Country Code		C	C
20	Primary Account Number Country Code		C	C
22	Point of Service Entry Mode		M	
23	Card Sequence Number		C	C
25	Point of Service Condition Code		C	C
26	Point of Service (POS) PIN Capture Code		C	
28	Amount, Transaction Fee		C	
32	Acquiring Institution Identification Code		M	M
33	Forwarding Institution Identification Code		C	
35	Track-2 Data		C	
37	Retrieval Reference Number		M	M
38	Authorisation Identification Response		C	
39	Response Code		M	M
41	Card Acceptor Terminal Identification		C	C
42	Card Acceptor Identification Code		C	C
43	Card Acceptor Name/Location		C	
45	Track-1 Data		C	
46	Proprietary Field 46		C	C
47	Proprietary Field 47		C	C
48	Proprietary Field 48		C	C
49	Currency Code, Transaction		M	M
51	Currency Code, Cardholder Billing		C	C
55	Smart Card Specific Data		C	C
100	Receiving Institution Identification Code		C	
102	Account Identification-1	C	C	
103	Account Identification-2	C	C	
104	Transaction Description	C	C	

Authorisation Reversal Messages (0420/0430)

Field Number and Name		0420	0430	
02	Primary Account Number (PAN)		M	M

⁵ For transit requests, this field may be omitted.
⁶ For transit requests, this field may be omitted.

03	Processing Code	M	M
04	Amount, Transaction	M	M
06	Amount, Cardholder Billing	C	C
07	Transmission Date and Time	M	M
10	Conversion Rate, Cardholder Billing	C	C
11	System Trace Audit Number	M	M
12	Time, Local Transaction	M ⁷	
13	Date, Local Transaction	M ⁸	
14	Date, Expiration	C	
18	Merchant Type	M	
19	Acquiring Institution Country Code	C	C
20	Primary Account Number Country Code	C	C
22	Point of Service Entry Mode	M	
23	Card Sequence Number	C	C
25	Point of Service Condition Code	C	C
28	Amount, Transaction Fee	C	
32	Acquiring Institution Identification Code	M	M
33	Forwarding Institution Identification Code	C	
37	Retrieval Reference Number	M	M
38	Authorisation Identification Response	C	
39	Response Code	C	M
41	Card Acceptor Terminal Identification	C	C
42	Card Acceptor Identification Code	C	C
43	Card Acceptor Name/Location	C	
46	Proprietary Field 46	C	C
47	Proprietary Field 47	C	C
48	Proprietary Field 48	C	C
49	Currency Code, Transaction	M	M
51	Currency Code, Cardholder Billing	C	C
90	Original Data Elements	C	C
95	Replacement Amounts	C	C
100	Receiving Institution Identification Code	C	
102	Account Identification-1	C	C
103	Account Identification-2	C	C
104	Transaction Description	C	C
112	Proprietary Field 112	C	C

Matching Principles

There are two types of matching. The first one is matching a response message to its request. The second is linking messages into a transaction chain (for example, authorisation and reversal messages).

For the first purpose, DE11 is used. DE11 must always be present in response messages of all types. It must be equal to the value of the request message's DE11.

To link messages into a transaction chain, a combination of DE32 and DE37 is used in the WAY 4 host-to-host protocol. The combination of field 32 and field 37 must be unique for each transaction (see the description in "DE37 Retrieval Reference Number").

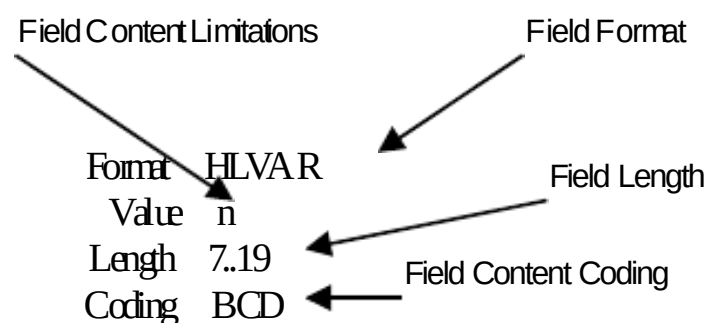
⁷ For transit requests, this field may be omitted.
⁸ For transit requests, this field may be omitted.

For transit MasterCard transactions, uniqueness of the field 32 and field 37 combination might be violated. In that case, the MasterCard STR matching scheme can be used. To enable the STR matching scheme, a transaction source channel must be present in the request and response for MasterCard traffic (see tag 951 of field 47).

Data Elements

Interpreting Data Elements Format

Field Format Example



Field Format

FIXED – The field has a fixed length. The length is specified in the Length field. The length of the field is expressed in data positions. It is not always in bytes. See "Field Length".

HLVAR – The first byte of the field contains the length of the variable data element that follows. The length is in the BCD coding. The length of the field is expressed in data positions. It is not always in bytes. See "Field Length". For example, the hex string 0x'06123456' contains a field of length 6 with the "123456" value in the BCD coding.

HLLVAR – The first two bytes of the field contain the length of the variable data element that follows. The length is in the BCD coding. The length of the field is expressed in data positions. It is not always in bytes. See "Field Length". For example, the hex string 0x'0006123456' contains a field of length 6 with the "123456" value in the BCD coding.

Field Content Limitations

Field content may be limited to a specified set of symbols. The following groups of symbols are possible:

- a – alphabetic characters
- n – numeric characters
- an – alphabetic and numeric characters
- as – alphabetic and special characters
- ns – numeric and special characters
- ans – alphabetic, numeric and special characters
- z – symbols that can be encoded on Track II
- zpos – symbols that can be encoded on Track II with '=' as a track separator

If none of the above groups is specified, then there are no limitations on field content.

Field Length

Interpretation of a field length depends on a field format and coding. For the fixed length format, the field length is constant. For the variable length formats, the length can be specified as a range of possible length values or a maximum length value. The length is expressed in data positions. For different field coding, data positions are different. For the BIT coding, the data position is a bit. For the BIN coding, the data position is a byte. For the EBCDIC and ASCII coding, the data position is a character. For the BCD coding, the data position is a nibble (a decimal digit expressed as 4 bits).

Field Content Coding

BCD	– Binary Coded Decimal	⁹
EBCDIC	– EBCDIC text data	
ASCII	– ASCII text data	
BIT	– Bit string in the binary format	
BIN	– Binary Data	

DE02 Primary Account Number (PAN)

Format	HLVAR
Value	n
Length	7..19
Coding	BCD

PAN is a series of digits used to identify a customer account or relationship. This is a card account number of up to 19 numeric digits. The length specifies the number of digits in the account number.

DE03 Processing Code

Format	FIXED
Value	n
Length	6
Coding	BCD

A code that identifies (1) the cardholder transaction type and (2) the cardholder account types, if any, affected by the transaction. This is a fixed-length field for three data elements, as follows. The codes are shown in the “Valid Values” section.

⁹ Standard BCD coding assumes leading 0 in the first position for odd number of coded data positions.

Positions1	-2Transaction Type	A two-digit code identifying the cardholder transaction type or the processing centre function being processed. Possible values are: "00" Goods/Service Purchase "01" ATM Withdrawal "11" Unique (Quasi-Cash) "17" or "01" Cash Advance "30" Balance Inquiry
Positions3	-4Account Type(From):	A two-digit code identifying the account type affected by this transaction. (For domestic account transfers, the account <i>from</i> which an account transfer is made) Possible values are: "00" Default Account "10" Savings Account "20" Checking Account "30" Credit Card Account
Positions5	-6Account Type(to):	Not applicable. (For domestic account transfers, a two-digit code identifying the account type <i>to</i> which an account transfer is made.)

DE04 Amount, Transaction

Format FIXED
Value n
Length 12
Coding BCD

The amount of funds requested by the cardholder expressed in minimal units of the local currency (specified by the currency code in field 49). No decimal point appears in this field; the decimal place is assumed based on the currency. For Balance Inquiry, the value of this field in request and response messages must be equal to 0.

DE06 Amount, Cardholder Billing

Format FIXED
Value N
Length 12
Coding BCD

This field contains the transaction amount (DE04) converted into the currency used to bill the cardholder's account. The conversion rate is in DE10. No

¹⁰ Unique transactions from WAY4 might have processing code '00' as well as '11' for transit from VISA/MC.
¹¹ The processing code for Cash Advance depends on the source of a transaction.

decimal point appears in this field; the decimal place is implied based on the currency. If this field is present, the following fields are also required:

Conversion Rate, Cardholder Billing" and "DE51 Currency Code, Cardholder Billing" .

"DE10

DE07 Transmission Date and Time

Format FIXED

Value n

Length 10

Coding BCD

The date and time in GMT (Greenwich Mean Time) when the request or advice is transmitted through the host-to-host connection. This field indicates when a transaction is sent to a host-to-host connection. The sender of the transaction enters a new (GMT) date and time with each request. The receiving member saves the field and returns it in the response message.

Valid Values

The value must be a valid date and time in the "MMDDhhmmss" format, where MM must be 01 – 12, DD must be 01 – 31, hh must be 00 – 23, mm must be 00 – 59, ss must be 00 – 59

DE10 Conversion Rate, Cardholder Billing

Format FIXED

Value n

Length 8

Coding BCD

This field contains the rate used to convert the transaction amount (DE04) into the cardholder billing amount (DE06). The DE04 amount multiplied by this rate equals the DE06 amount. The left-most digit denotes the number of positions the decimal separator will be moved from the right (allowable values are 0 to 9). Positions 2 – 8 of the field are the actual rate. For example: 69985022=9.985022.

DE11 System Trace Audit Number

Format FIXED

Value n

Length 6

Coding BCD

A number assigned by the originator of the request message. The same value of the System Trace Audit Number (STAN) must be used for the response

message. The STAN of the request message must be as unique as possible. A good assumption is that it is unique during 24 hours.

DE12 Time, Local Transaction

Format FIXED
Value N
Length 6
Coding BCD

This field contains the time the transaction took place expressed in the local time of the card acceptor location. The date is in the "hhmmss" format where
hh=hours
mm=minutes
ss=seconds

DE13 Date, Local Transaction

Format FIXED
Value n
Length 4
Coding BCD

This field contains the month and day on which the cardholder originated the transaction. For recurring payments, this field contains the cardholder requested payment date. The date is in the "mmdd" format where
mm=month
dd=day

DE14 Date, Expiration

Format FIXED
Value n
Length 4
Coding BCD

The year and month after which the card expires. The date is in the "yyymm" numeric format, where yy=year (00 – 99) and mm=month (01 – 12). DE14 is mandatory for all card-present requests and for all cases when the true expiration date is known.

DE18 Merchant Type

Format FIXED
Value N
Length 4
Coding BCD

A code describing the merchant's type of business product or service. See ISO 8583 standard documentation for the list of valid categories. -

DE19 Acquiring Institution Country Code

Format FIXED
Value n
Length 3
Coding BCD

This code identifies the country of the acquiring institution for the transaction terminal. Valid values for this field are the ISO numeric country codes.

DE20 Primary Account Number Country Code

Format FIXED
Value n
Length 3
Coding BCD

A code identifying the country where the card issuer institution is located. This field is used for transactions with PAN starting with "59".

DE22 Point of Service Entry Mode

Format FIXED
Value n
Length 3
Coding BCD

Point of Service Entry Mode indicates the method by which the Primary Account Number (PAN) was captured at the point of service and, where applicable, the PIN entry capabilities of the terminal used.

Positions 1-2, PAN and Date Entry Mode: A 2-digit code that identifies the actual method used to enter the cardholder account number. This code specifies whether the entire magnetic stripe is included in an authorisation request.

"00" Unknown or terminal not used.

"01" Manual (key entry).

- "02" Magnetic stripe read.
- "05" Integrated circuit card read; card data reliable.
- "06" PAN auto-entry via a chip MasterCard PayPass Mapping Service applied when DE 22, subfield 1 contains a value of 05.
- "07" PAN auto-entry via a contactless M/Chip.
- "08" Contactless M/Chip MasterCard PayPass Mapping Service applied when DE 22, subfield 1 contains a value of 07.
- "79" Chip card at chip-capable terminal was unable to process the transaction using chip technology. The transaction proceeds as fallback to a magnetic stripe read transaction, but the terminal is unable to send the transaction online. The transaction is subsequently authorized by the Voice Center, where the PAN is keyed in manually by the acquirer.
- "80" PAN auto-entry via magnetic stripe – the full track data has been read and transmitted in Track-2 Data (DE 035) without alteration or truncation. This mode is used as fallback to PAN auto-entry via ICC when the IC could not be used.
- "81" PAN auto-entry via electronic commerce.
- "82" PAN Auto Entry via Server (issuer, acquirer, or third party vendor system).
- "90" Magnetic stripe read and exact content of Track 1 or Track 2 included (CVV check is possible).
- "91" PAN auto-entry with contactless Magnetic Stripe - The full Track data has been read from the data on the card and transmitted within the authorization request in Track 2 Data (DE 35) or Track 1 (DE 45) without alteration or truncation.
- "92" Contactless input, MasterCard PayPass Mapping Service applied when acquirer DE 22, subfield 1 = 91.
- "95" Integrated circuit card; card data may be unreliable.
- "96" Stored Value from Pre-Registered Checkout Service.

Position 3 PIN Entry Capability: A 1-digit code that identifies the terminal's capability of capturing PINs. This code does not necessarily mean that a PIN was entered or is included in this message. For example, the use of code 1 does not imply the presence of a PIN. If the code is 2 or 8, however, it is reasonable to assume that a PIN is not present.

- "0" Unknown
- "1" Terminal can accept PINs
- "2" Terminal cannot accept PINs
- "8" The terminal PIN pad is down
- "9" Reserved for future use

In addition to the listed codes, any codes provided by payment systems might be present in the W A Y 4 host-to-host interface for transit messages

DE23 Card Sequence Number

Format FIXED
Value n
Length 3
Coding BCD

The Card Sequence number is a mandatory field for ICC-originated transactions. This data is provided by smart card - EMV Tag 5F34.

Field 023 contains the Card Sequence Number in the range '000' to '099', rightjustified and left-padded with zeros.

DE25 Point of Service Condition Code

Format FIXED
Value N
Length 2
Coding BCD

A code identifying transaction conditions at the point of service, thus, in many cases, identifying a type of an original or subsequent transaction. The following values are possible:

- "00" Normal transaction of this type
- "01" Cardholder not present
- "02" Unattended acceptance terminal, customer operated (for example, A TM, Automated Dispensing Machine)
- "03" Merchant suspicious of transaction (or card)
- "05" Customer present, Card Not Present
- "08" Mail/telephone order (includes VisaPhone and recurring transactions)
- "10" Customer identity verified (used for Check Acceptance only)
- "51" Request for account number verification without authorisation, or request for account number verification and address verification without authorisation
- "55" Request for telecode verification without authorisation
- "59" E-commerce request (through public network)
- "71" Card present, magnetic stripe cannot be read (key-entered)

For Balance Inquiry transactions, this field may be omitted. For Mail Order transactions, it must be present and equal to "08".

DE26 Point of Service (POS) PIN Capture Code

Format FIXED
Value n
Length 2
Coding BCD

The number of PIN characters that the acquiring terminal is capable of accepting. If the field is omitted, the default value "12" is assumed.

DE28 Amount, Transaction Fee

Format FIXED
Value an
Length 9
Coding ASCII

Field 28 is the fee charged (for example, by the acquirer) for transaction activity in the currency of Field 4 (Amount, Transaction).

The fee amount is included in field 4. The 1 AN prefix is "C" for credit or "D" for debit, as appropriate.

This fee is in the same currency as that used in Field 4 — Amount, Transaction.
The currency code in Field 49 — Currency Code, Transaction applies. The number of decimal places assumed for this field depends on the currency. If that currency is defined with three decimal places, the last digit of field 28 must be zero.

DE32 Acquiring Institution Identification Code

Format HVAR
Value n
Length 6..11
Coding BCD

This code identifies the financial institution acting as the acquirer of this cardholder transaction. The acquirer is a member or system user that installed the terminal. The ID can be a Visa BIN, a MasterCard BIN, or another code that identifies the financial institution or branch within WAY 4. BINs are usually six digits, but the code may be up to 11 digits long. The value length specifies the number of digits in the ID code.

DE33 Forwarding Institution Identification Code

Format HLVAR
Value n
Length 6..11
Coding BCD

This is a code that identifies the institution that forwards the request.

DE35 Track-2 Data

Format HLVAR
Value z
Length ..37
Coding BCD

For magnetic stripe-read transactions, this data element carries the information encoded on track 2 of the magnetic stripe of the card, as defined in ISO 7813. For chip transactions, this data element carries data read from the chip as "Track 2 Equivalent Data" (EMV tag 57), which is then treated in the same way as magnetic stripe data. The account number in DE 002, Primary Account Number, must match the account number encoded on track 2 and provided in this field.

For magnetic stripe-read transactions, Track-2 Data may be provided along with Track-1 Data (DE045) as well as alone.

The Track 2 delimiter/separator character (≡) must be encoded as 0xD (binary 1101).

DE37 Retrieval Reference Number

Format FIXED
Value an
Length 12
Coding ASCII

A number that is used with other data elements as a key to identify and track all messages related to a given cardholder transaction. The retrieval reference number is a key data element for matching a message to others within a given transaction set. The reference number must be the same in all messages for the set. For example, a new retrieval number is assigned when a financial transaction is processed. The same number appears in all related messages: response, advice, and reversal.

The combination of DE37 and field DE32 must be unique to link the messages belonging to the same transaction into a message chain. The valid value for this field is YJJXXNNNNNN, where Y is the last digit of the year,

JJJ is a Julian date,
XX is a unique interface ID assigned by OpenWay, and
NNNNNN is a counter unique during the day.

For transit messages coming from payment systems like VISA or MasterCard,
the value of the DE37 may violate the rules described above.

DE38 Authorisation Identification Response

Format FIXED
Value an
Length 6
Coding ASCII

An authorisation code provided by the issuer when a transaction is approved, or
a no-reason-to-decline code provided for successful verifications.

DE39 Response Code

Format FIXED
Value an
Length 2
Coding ASCII

A code that defines the response to a request or the message disposition. Code
00 indicates both *approval* (a positive authorisation decision) and *acceptance*
(acknowledgement of receipt of a transaction or a message).

This is a list of response codes returned by WAY 4¹². All codes are divided into
three categories: Approve, Decline and Pickup.

00	Approved	Approve
01	Refer to Card Issuer	Decline
02	Refer to Card Issuer, special condition	Decline
03	Invalid Merchant	Decline
04	Pick up card	Pickup
05	Do not honour	Decline
06	Error	Decline
07	Pick up card, special condition	Pickup
09	Request in progress	Decline
12	Invalid Transaction	Decline
13	Invalid Amount	Decline
14	Invalid card number	Decline
19	Re-enter transaction	Decline
21	No action taken	Decline
30	Format Error	Decline
41	Lost card pick up	Pickup
43	Stolen card pick up	Pickup
51	Not sufficient funds	Decline
52	No checking account	Decline
53	No savings account	Decline
54	Expired card	Decline
55	PIN incorrect	Decline
57	Transaction not allowed for cardholder	Decline
58	Transaction not allowed for merchant	Decline
61	Exceeds withdrawal amount limit	Decline
62	Restricted card	Decline

¹² The list of response codes may vary depending on WAY 4 setup.

63	Security violation	Decline
65	Activity count limit exceeded	Decline
75	PIN tries exceeded	Decline
77	Inconsistent with original	Decline
78	No account	Decline
84	Pre-authorisation time too great	Decline
86	Cannot verify PIN	Decline
91	Issuer unavailable	Decline
92	Invalid receiving institution id	Decline
93	Transaction violates law	Decline
94	Duplicate transaction	Decline
96	System malfunction	Decline

Any other codes not listed here must be treated as decline response codes.

In addition to the listed codes, any codes provided by payment systems such as VISA or MasterCard might be present in the WAY 4 host-to-host interface for transit messages.

If Member receives request originated by VISA or MasterCard, it should respond with response code according to particular payment system specification. Member should do it also in the case if this response code is not listed in standard WAY 4 response codes table shown above.

In 0420 reversal messages, DE39 might be used to carry an action code. If DE39 is present in an 0420 message, the following action codes are only permitted.

06	Error	
	Indicates that the reason for the reversal was due to an error in a previous message.	
17	Customer cancellation	
	The cardholder aborted the transaction after an authorization request had been sent.	
32	Completed partially	
	Used in reversal messages to indicate that a transaction was completed, but for an amount less than that given in the original 0100 request	
68	Response received too late	
	Used to indicate the reason for sending a reversal authorisation response was not returned within the time limit set by the acquirer.	— an
96	System malfunction	
	The message could not be successfully processed due to a system malfunction (see also Note below)	

DE41 Card Acceptor Terminal Id

Format FIXED
Value Ans
Length 8
Coding ASCII

A code that uniquely identifies a particular terminal at the card acceptor's location. An identification code of less than eight positions must be left-justified and the remainder of the field space-filled. This field is mandatory if the terminal id is available.

DE42 Card Acceptor Identification Code

Format FIXED
 Value Ans
 Length 15
 Coding ASCII

A code that uniquely identifies a particular card acceptor (i.e. a merchant or a bank). A Card Acceptor Identification Code is required in all authorisation requests and related advice messages. It is a string of 15 characters assigned by the acquirer and padded as required. For Retail transactions, DE 042 must contain a Merchant Identification that uniquely identifies the merchant within the acquiring institution. For ATM and bank branch transactions, DE 042 must contain a code that uniquely identifies the bank (for example, its name or national code number).

DE43 Card Acceptor Name/Location

Format FIXED
 Value ans
 Length 40
 Coding ASCII

The name and location of the card acceptor. This field is required in authorisation requests, advice messages and related reversal requests for all card-read transactions. It contains the information necessary for printing on customer account statements and on credit card billing statements.

Positions 1 – 25 Terminal Location:

POS: Merchant name as known to the cardholder; or, for original credit money transfer transactions, the sender name or transaction-related data.

ATM: The ATM location, branch number, or street address.

Positions 26 – 38 City Name:

POS: City where the customer transaction occurs.

ATM: City where the ATM is located.

Positions 39 – 40 Country Code The two-character ISO alpha country code for the country where the terminal is located. Country codes must be upper case.

DE 44 Additional Response Data

Format HLVAR
Value Ans
Length 0.25
Coding ASCII

Field 44 contains miscellaneous response message data.

DE 45 Track-1 Data

Format HLVAR
Value Ans
Length .76
Coding ASCII

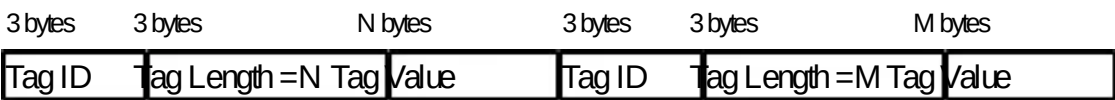
This data element carries the information encoded on track 1 of the magnetic stripe of the card, as defined in ISO 7813. For magnetic stripe-read transactions, if track 2 is not provided, this field must be present and contain the data encoded on track 1 of the magnetic stripe.

For magnetic stripe-read transactions, Track-1 Data may be provided along with Track-2 Data (DE035) as well as alone.

DE 46 Proprietary Field 46

Format HLLVAR
Value ans
Length 0..999
Coding ASCII

This field is used to carry WAY 4 internal information. There is a number of tags packed in field 46. The tags have the TLV structure. It means they are repeated groups of Tag IDs, Tag Lengths and Tag Values.

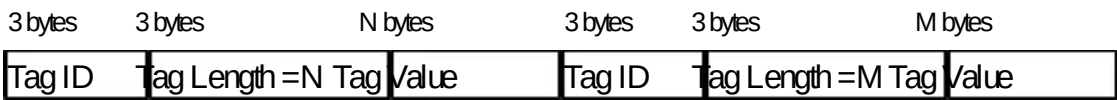


A Tag ID is a 3-digit decimal string from "701" to "799". A Tag Length is a 3-digit decimal string from "001" to "999". A Tag Value may contain any printable data.

DE47 Proprietary Field 47

Format HLLVAR
Value ars
Length 0..999
Coding ASCII

This field is used to carry WA Y 4 internal information. There is a number of tags packed in field 47. The tags have the TL V structure. It means they are repeated groups of Tag IDs, Tag Lengths and Tag Values.

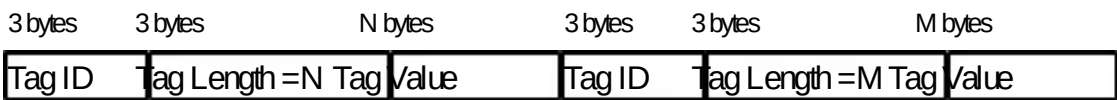


A Tag ID is a 3-digit decimal string from "901" to "999". A Tag Length is a 3-digit decimal string from "001" to "993". A Tag Value may contain any printable data. For example, field value "902001X905002Y Y " contains tag "902" with value "X" and tag "905" with value "Y Y".

DE48 Proprietary Field 48

Format HLLVAR
Value ars
Length 0..999
Coding ASCII

This field is used to carry WA Y 4 internal information. There is a number of tags packed in field 48. The tags have the TL V structure. It means they are repeated groups of Tag IDs, Tag Lengths and Tag Values.



A Tag ID is a 3-digit decimal string from "801" to "899". A Tag Length is a 3-digit decimal string from "001" to "993". A Tag Value may contain any printable data.

DE49 Currency Code, Transaction

Format FIXED
Value N
Length 3
Coding BCD

This field contains a 3-digit numeric code that identifies the transaction currency in the following amount fields:

DE04 — Amount, Transaction, and
DE95.1 — Actual Amount, Transaction.

This code is used to determine the number of decimal places in the above amount fields.

This field is used in any message related to a cardholder transaction that contains one or more of the amount fields listed above, even when the amount is zero. This field is also used in balance inquiry requests, even when the value of field 4 equals zero. This code specifies the currency in which the acquirer wants the balance amount to be expressed.

DE51 Currency Code, Cardholder Billing

Format FIXED
Value n
Length 3
Coding BCD

This field contains a 3-digit numeric code that identifies the currency used by the issuer to bill the cardholder's account in the following amount field:

DE06 — Amount, Cardholder Billing, and
DE95.3 — Actual Amount, Cardholder Billing.

This code is used to determine the number of decimal places in the above amount field.

DE52 Personal Identification Number (PIN) Data

Format FIXED
Value
Length 64
Coding BIT

A binary data block containing either a derived encrypted value calculated from the PIN entered by the customer at the point of service, or the PIN validation results in the case of a pre-validated PIN.

If this field is present, DE53 — Security Related Control Information must also be present. The format of this field must conform to the PIN Block Format Code in DE53 — Security Related Control Information.

It is not used in reversal requests or advices, or in any responses.

For pre-validated PINs, the first nibble should equal binary 1111. The second nibble should equal binary 0000. The third one should carry a validation result, a four-bit binary number with permissible values of:

"0000" (=0) PIN validation impossible for technical reasons

"0001" (=1) PIN validated as correct

"0010" (=2) PIN validated as incorrect

DE53 Security Related Control Information

Format FIXED
Value N
Length 16
Coding BCD

Security information used to identify the Security Format, PIN Block Format, encryption algorithm, and PIN Key Index used in the PIN Data element (DE 052).

Positions 1 – 2, Security Format Code: This code defines the security technique used.

"96" Pre-Validated PIN

"98" Zone encryption

Positions 3 – 4, PIN Encryption Algorithm Identifier: This code defines the encryption technique used.

"00" PIN in clear (but ONLY used for pre-validated PINs)

"01" PIN block encrypted using the DES algorithm

Positions 5 – 6, PIN Block Format Code: This code defines the format of DE52.

"01" ANSI format

"25" Pre-validated PIN format

Positions 7 – 16, Not used: This subelement is not used and normally should be filled in with zeroes.

PIN Encryption Algorithm Identifier "00" (no encryption) is ONLY valid with PIN Format Code "25" (pre-validated PIN).

DE54 Additional Amounts

Format HLLVAR
Value ars
Length 20..120
Coding ASCII

This field contains account balance information. Balance information is for the cardholder's benefit. This field is used to return account balance information.

After the length subfield, there are six possible sets of the following subfields:

Positions 1 – 2, Account Type: A 2-digit code identifying the account type affected by the balance inquiry. Valid codes are:

"00" Not Applicable or Not Specified

"10" Savings Account

"20" Checking Account

"30" Credit Card Account

Positions 3 – 4, Amount Type: A 2-digit code describing the use of the amount. Valid codes are:

"01" Current ledger (posted) balance

"02" Current available balance (typically, ledger balance minus outstanding authorisations)

"90" Available Credit

"91" Credit Limit

Positions 5 – 7, Currency Code A 3-digit code that defines the currency used in Positions 9 – 20.

Position 8, Amount, Sign A 1-digit code that defines the value of the amount as either positive or negative.

"C" Positive balance

"D" Negative balance

Positions 9 – 20, Amount A twelve-character amount that is right-justified and contains leading zeros. The amount also includes an implied decimal relative to the currency code specified in positions 5 – 7.

DE 55 Smart Card Specific Data

Format HLLVAR

Length .255

Coding BIN

Smart Card Specific Data (DE 055) is used to transport chip-specific data over the payment network. It contains various data captured by the chip terminal as part of a chip transaction. DE 055 is used only for Chip Full Grade transactions.

DE 70 Network Management Information Code

Format FIXED

Value n

Length 3

Coding BCD

A code used to define the purpose of a Network Management Request (0800) message – whether for sign-in, sign-off or an echo test message. Network Management Request (0800) messages are used to control the status of an application session between hosts. The Network Management Information Code in DE 070 indicates the principal reason for sending a 0800 request message. The WAY 4 host-to-host protocol supports the following Network Management Information Codes:

"001" Indicates that the 0800 message is a Sign-In Request

"002" Indicates that the 0800 message is a Sign-Off Request

"301" Indicates that the 0800 message is an Echo-Test Message

DE 90 Original Data Elements

Format FIXED

Value n

Length 42

Coding BCD

DE90 contains information for tracking the current message back to prior messages for the same cardholder transaction, for example, a reversal to an original request. This field is fixed-length with five subfields.

Original Message Type (subfield 901): This subfield contains the 4-digit message type identifier from the original message for the transaction being reversed.

Original Trace Number (subfield 902): This subfield contains the 6-digit trace number from field 11 of the original message.

Original Transmission Date and Time (subfield 903): This subfield contains the 10-digit transmission date and time from field 7 of the original message.

Original Acquirer ID (subfield 904): This subfield contains 11 positions for the acquiring institution ID from field 32 of the original message, right-justified, with leading zero fill.

Original Forwarding Institution ID (subfield 905): This subfield contains 11 positions for the forwarding institution ID from DE33 of the original message, which is right-justified, with leading zero fill.

DE 91 File Update Code

Format FIXED

Value AN

Length 1

Coding ASCII

DE 91 is used in 03xx Stop List messages.

The following values are possible:

"1" Add

"2" Update

"3" Delete

"4" Replace (VISA only)

"5" Inquire (VISA, MasterCard Only).

DE 95 Replacement Amounts

Format FIXED
Value N
Length 42
Coding BCD

DE95 contains the corrected amount of a transaction in a partial reversal. This is defined as a fixed-length field with four subfields, but only the first subfield is used. Two other subfields may contain data received from VISA or MasterCard.

Actual Amount, Transaction (subfield 951): This 12-position field is used for the corrected, actual amount of the cardholder's transaction, in the transaction currency. The value is right justified, with leading zero fill.

Actual Amount, Settlement (subfield 952): This 12-position field is used for the corrected, actual amount of the customer's transaction, in the settlement currency. The field, which is unused, must be zero-filled.

Actual Amount, Cardholder Billing (subfield 953): This 12-position field is used for the corrected, actual cardholder billing amount of the customer's transaction, in the cardholder billing currency. The field, which is unused, must be zero-filled.

Zero fill (subfield 954): This 6-position field is unused and must be zero-filled.

This subelement definitions differ slightly from the ISO subelement definitions for this dataelement.

DE100 Receiving Institution Identification Code

Format HLVAR
Value n
Length ..11
Coding BCD

A code that identifies the institution that should receive a request or advice. This ID is used when it is not possible to route a message using the account number field in the message. When this field is included in a request or advice, it takes precedence over all account number fields for routing.

DE102 Account Identification-1

Format HLVAR
Value ans
Length 0..99
Coding ASCII

In cardholder transactions, a number that identifies an account or cardholder relationship.

DE 103 Account Identification-2

Format HLVAR
Value ans
Length 0..99
Coding ASCII

In cardholder transactions, a number that identifies an account or cardholder relationship.

DE 104 Transaction Description

Format HLVAR
Value ans
Length 0..99
Coding ASCII

This ISO-defined field contains additional billing or reporting information about transaction or message processing.

DE 112 Proprietary Field 112

Format HLLVAR
Length ..999
Coding BIN

This field is used to carry WAY 4 internal information. The field can consist of a set of the following sub-fields. Each sub-field is coded in BER-TLV. The format description is based on “EMV Integrated Circuit Card Specifications for Payment Systems Book 3 Application Specification”, Annex B Rules for BER-TLV Data Objects.

For convenience we included this description in the chapter “General Rules for BER-TLV Data Objects” below.

General Rules for BER-TLV Data Objects

As defined in ISO/IEC 8825, a BER-TLV data object consists of 2-3 consecutive fields:

- The tag field (T) consists of one or more consecutive bytes. It indicates a class, a type, and a number (see Table 1). The tag field of the data objects described in this specification is coded on one or two bytes.

- The length field (L) consists of one or more consecutive bytes. It indicates the length of the following field. The length field of the data objects described in this specification is coded on one or two bytes.
- The value field (V) indicates the value of the data object. If L = '00', the value field is not present.

A BER-TLV data object belongs to one of the following two categories:

- A primitive data object where the value field contains a data element for financial transaction interchange.
- A constructed data object where the value field contains one or more primitive or constructed data objects.

The coding of BER-TLV data objects is defined as follows.

Coding of the Tag Field of BER-TLV Data Objects

Table 1 Tag Field Structure (First Byte) BER-TLV

b8	b7	b6	b5	b4	b3	b2	b1		Meaning
0	0								Universal class
0	1								Application class
1	0								Context-specific class
1	1								Private class
		0							Primitive data object
		1							Constructed data object
				1	1	1	1	1	See subsequent bytes
Any other value <31							Tag number		

According to ISO/IEC 8825, Table 2 defines the coding rules of the subsequent bytes of a BER-TLV tag when tag numbers of the first byte equal '11111').

- 31 are used (that is, bits b5 - b1

Table 2 Tag Field Structure (Subsequent Bytes) BER-TLV

b8	b7	b6	b5	b4	b3	b2	b1		Meaning
1									Another byte follows
0									Last tag byte
Any value >0									(P art of) tag number

Coding of the Length Field of BER-TLV Data Objects

When bit b8 of the most significant byte of the length field is set to 0, the length field consists of only one byte. Bits b7 to b1 code the number of bytes of the value field. The length field is within the range 1 to 127.

When bit b8 of the most significant byte of the length field is set to 1, the subsequent bits b7 to b1 of the most significant byte code the number of subsequent bytes in the length field. The subsequent bytes code an integer representing the number of bytes in the value field. Two bytes are necessary to express up to 255 bytes in the value field.

Coding of the Value Field of Data Objects

A data element is the value field (V) of a primitive BER-TLV data object. A data element is the smallest data field that receives an identifier (a tag).

A primitive data object is structured as illustrated in Figure 1:

Figure 1: Primitive BER-TLV Data Object (Data Element)

Tag(T)	Length(L)	Value(V)
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A constructed BER-TLV data object consists of a tag, a length, and a value field composed of one or more BER-TLV data objects. A constructed data object is structured as illustrated in Figure 2:

Figure 2: Constructed BER-TLV Data Object

Tag(T)	Length(L)	Primitive or constructed BER-TLV data object number 1	...	Primitive or constructed BER-TLV data object number n
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Using of BER-TLV structures in OpenWay Host-to-host dialect

This format allows to transfer data of the arbitrary structure. The data structure and list of elements (tags names) to support current requirement to transfer non-standard information. New tags could be added to the list if the new requirements to transfer custom information arise. The tags names should be obtained from WAY 4.

BER-TLV Primitive tag format

All values of primitive tags should contain data type in the first byte. This byte is taken into account for the data element processing, conversion and its length calculation.

Primitive Tag Value

Value Type	Value
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Value Type can be one of the following:

- ‘D’ – BCD-packed numeric data with leading nonsignificant zeroes;
- ‘F’ – BCD-packed numeric data with leading significant zeroes, right padded by ‘F’ if necessary (odd count of data elements);
- ‘T’ – string of literals;
- ‘B’ – binary data in the form of hex-decimal string

BER-TLV Constructed Tag Format

Constructed tags have data type ‘B’ by default. This type should not be part of tag value as according to basic BER-TLV specification it can be derived from the tag name (see BER-TLV specification, chapter “Coding of the Tag Field of BER- TLV Data Objects”) .

NOTE: Any new TAG of any level should be assigned by OpenWay.

Example of BER-TLV Constructed Tag List (Level n)

Tag	Len	Value						Comment
EE	4C	Tag	Len	Type	Value			
		D1	08	54	4558414D504C45	Text tag		EXAMPLE
		FF01	1E		Tag	Len	Type	Value Sample
					D1	0E	54	435553544F4D5F444154415F41
								Text tag – CUSTOM DATA A
					C1	07	46	00000000500 ISO Amount
					C2	03	44	0643 ISO Currency Code
		FF02	1E		Tag	Len	Type	Value Sample
					D1	0E	54	435553544F4D5F444154415F42
								Text tag – CUSTOM DATA B
					C3	06	54	2D352E3230 Text Amount – ‘-5.20’
					C4	04	54	525552 Currency - RUR

DE122 Additional Record Data

Format HLLVAR
Value Ans
Length 0..999
Coding ASCII

Additional Record Data (DE 122) is a free-format variable length data element used for transmitting file record data in various message types.